

PULSE ARTWORK STUDIO AI

Product Requirements Document

Version 1.0 • MVP / Genesis

Create • Enhance • Print Smarter

Product Owner: Pulse Machinery

Platform: Responsive Web Application

Frontend: Flutter

Document Status: Foundation Specification

Document Purpose

This document is the single source of truth for the first commercial release of Pulse Artwork Studio AI. It defines the product vision, users, functional requirements, MVP scope, experience principles, quality requirements, and recommended technology direction. Detailed implementation decisions will be captured in the subsequent System Architecture & Technical Design Specification.

Product Snapshot

Item	Decision
Product	Pulse Artwork Studio AI
Business	Pulse Machinery
Primary market	DTF print shops and garment decorators
Platform	Responsive web application
Frontend	Flutter Web
Backend	Python + FastAPI
Database	PostgreSQL
Async processing	Redis + Celery workers
Storage	S3-compatible object storage
MVP focus	Artwork preparation and DTF quality assistance

1. Product Overview

Overview

Pulse Artwork Studio AI is a cloud-enabled, AI-powered artwork preparation platform designed specifically for DTF printing and garment-decoration businesses. It consolidates common artwork cleanup and preparation tasks into one workflow.

Core workflow

Upload → Analyze → Improve → Prepare → Preview → Export

Primary value

Reduce artwork-preparation time, lower technical barriers for beginners, improve consistency, and help print operators identify common DTF artwork problems before production.

2. Product Vision

Vision statement

To become a leading AI-powered artwork preparation platform for garment decoration and digital printing.

Long-term direction

The architecture should allow future expansion into UV DTF, DTG, sublimation, embroidery, vinyl, screen printing, UV printing, laser engraving, and signage. DTF remains the Version 1 priority.

3. Problem Statement

Current pain

DTF businesses frequently receive low-resolution, blurry, poorly cropped, incorrectly sized, non-transparent, over-detailed, or otherwise unsuitable artwork.

Current workflow problem

Users may need several separate applications for background removal, enhancement, vectorization, effects, layout, and final preparation.

Business impact

This increases software costs, operator training, processing time, customer-service effort, and the risk of inconsistent print results.

4. Target Users

DTF print shop

Produces T-shirts, hoodies, workwear, sportswear, bags, caps, and promotional apparel. Needs fast, reliable and repeatable preparation.

Beginner

A new printer owner who may not understand DPI, transparency, vector graphics, underbases, halftones, or print limitations. The application must explain issues in plain language.

Professional designer

Needs precise dimensions, vector output, effects, advanced controls, and predictable exports.

Pulse Machinery customer

Future dedicated experience with printer profiles, training, support and recommended artwork settings.

5. Product Principles

Simple

A beginner should be able to upload artwork and produce usable output without formal graphics training.

Professional

Advanced users must retain meaningful control over processing and output.

DTF first

Processing recommendations should consider the realities of DTF production.

AI assisted, not AI controlled

AI recommendations must be reviewable and overridable.

Non-destructive

Original artwork must never be permanently modified.

6. Core User Journey

Journey

Login → Dashboard → New Project → Upload Artwork → AI Artwork Analysis → Fix Automatically or Edit Manually → Process → Preview → DTF Check → Export → Reopen Project Later.

7. Dashboard Requirements

Primary actions

New Project; Upload Artwork; AI Artwork Inspector; Gang Sheet; Recent Projects; Templates; AI Assistant.

Dashboard information

Recent projects, processed artwork count, storage usage, subscription status, recent exports, and future AI-credit usage.

8. Artwork Editor

Workspace

The editor is the heart of the application and must support an artwork canvas, tool navigation, properties, zoom/pan, undo/redo, and status information.

Desktop layout

Left: AI and creative tools. Centre: artwork canvas. Right: properties and print information. Top: file/edit/AI/effects/DTF/view/export navigation. Bottom: zoom, position and DTF quality status.

Responsive behavior

On tablets and phones, panels should collapse into drawers, tabs, bottom sheets, or contextual controls rather than simply shrinking the desktop interface.

9. AI Artwork Inspector

Purpose

Analyze uploaded artwork and identify likely preparation or printing issues.

Potential checks

Resolution too low; background detected; image blur; color concerns; transparency; small text; thin lines; potential clipping; unsuitable dimensions.

Output

A clear quality score, detected issues, explanations, recommended actions, and a one-click Fix All option where technically safe.

10. AI Enhancement

Standard tools

Sharpen, denoise, deblur, contrast, brightness, saturation, and color correction.

AI tools

AI upscaling, detail reconstruction, artifact removal, intelligent sharpening, and edge enhancement.

Upscaling

2×, 4× and 8× options, with before/after comparison.

11. Background Removal

Capabilities

Automatic segmentation, transparent PNG output, edge refinement, white-background removal, black-background removal, selected-color removal, and manual brush refinement.

Requirement

Original artwork must remain unchanged.

12. Vector Conversion

Inputs

JPG, JPEG, PNG, and WEBP.

Outputs

SVG and PDF in MVP; EPS is a future capability.

Controls

Low/medium/high/maximum vectorization quality, original or reduced colors, custom palette, black-and-white, curve smoothing, small-object removal, path simplification, and noise removal.

13. Halftone Generator

Styles

Classic dots, CMYK, gradient, vintage, newspaper, line halftone, and geometric.

Controls

Dot size, frequency, angle, contrast, density, and shape.

UX

Preview should update dynamically as practical.

14. Embroidery Effect

MVP definition

Generate a realistic visual embroidery simulation rather than a machine-ready embroidery file.

Effects

Satin stitch, fill stitch, running stitch, thread texture, stitch direction, thread density, and 3D/puff appearance.

Scope protection

DST/PES embroidery digitization is explicitly outside MVP and should be a future module.

15. DTF Preparation

Controls

Width, height, DPI, physical print size, transparency, background, rotation, flip, mirror, scale, crop, and canvas settings.

Print checks

Resolution, fine lines, small text, transparency, dimensions, and potential clipping.

16. White Underbase Preview

Modes

Full Color, CMYK, White Underbase, and CMYK + White.

Important limitation

This is a visual simulation and must not be presented as a replacement for the RIP software's actual underbase generation.

17. Gang Sheet Builder

Planned capability

Add multiple artworks, automatically arrange them, define sheet width/height, set spacing, rotate, duplicate, and automatically nest designs.

Presets

30 cm, 33 cm, 60 cm, and custom sizes.

Release status

Should follow the core artwork editor in the MVP sequence if schedule pressure requires prioritization.

18. Export System

MVP outputs

Transparent PNG, SVG, and PDF.

PNG requirements

Support high-resolution output and 300 DPI metadata.

Future outputs

TIFF, EPS, PSD, and additional production formats.

19. Project System

Project contents

Original artwork, processed artwork, effects/settings, versions, and exports.

Versioning

Users must be able to reopen a project and continue editing without losing the original asset.

20. Authentication

MVP architecture

Email/password authentication with secure sessions; password reset and email verification should be supported.

Future

Google, Microsoft, and Apple sign-in.

21. Licensing & Subscription

Planned tiers

Free, Professional, Business, and Enterprise.

Architecture requirement

Licensing and entitlements must be designed from the beginning even if billing is activated after the core MVP.

22. Recommended Technology Stack

Frontend

Flutter Web for responsive UI, shared component architecture, future mobile expansion, navigation, project management, canvas interaction, and API communication.

Backend

Python + FastAPI. This is the preferred backend because image processing, computer vision, vectorization, and machine-learning workloads have a strong Python ecosystem.

Database

PostgreSQL for users, projects, metadata, jobs, subscriptions, settings, usage data, and audit records.

Queue/cache

Redis for queues, caching, rate limiting, and temporary state.

Workers

Celery for computationally expensive asynchronous processing.

Image processing

Pillow + OpenCV + NumPy, supplemented by ImageMagick and vector-processing tools where appropriate.

Vectorization

Potrace + SVG tooling for raster-to-vector workflows.

Storage

S3-compatible object storage for originals, processed files, previews and exports; PostgreSQL should store metadata, not large binary assets.

Deployment

Docker-based services with CI/CD through GitHub Actions and secure environment-based secrets.

23. Recommended Architecture

Principle

Flutter should be the presentation layer; Python should be the intelligence and processing layer.

Pipeline

Flutter → HTTPS/API → FastAPI → job queue → AI/image worker → object storage → job status → Flutter.

Scalability

AI workers should be independently scalable and replaceable so models can improve without rewriting the application.

24. Security Requirements

Required

HTTPS, secure authentication, password hashing, protected sessions/JWT architecture, API rate limiting, file validation, malware/file-type checks, signed download URLs, access control, user/project isolation, and secure server-side API keys.

Critical rule

AI provider secrets and private API keys must never be exposed to the Flutter client.

25. Performance Requirements

Targets

Initial application load under approximately 3 seconds under normal conditions; responsive interactions targeted below 100 ms where practical; visible upload and processing progress; no UI freezing during AI processing.

Large files

Large-image and AI processing must be asynchronous.

26. Error Handling

User experience

Technical failures must be translated into understandable messages with recovery actions such as Try Again, Upload Another File, or Contact Support.

Logging

Detailed technical errors should be captured in server-side monitoring rather than exposed to end users.

27. Accessibility

Requirements

Keyboard navigation, accessible labels, sufficient contrast, screen-reader-friendly controls, scalable text, and touch-friendly controls.

28. Analytics & Privacy

Metrics

Most-used tools, processing failures, processing time, export formats, effect usage, retention, and subscription conversion.

Privacy

Customer artwork should not be used for analytics, training, or model improvement without appropriate consent and an explicit product policy.

29. MVP Scope

MUST HAVE

Authentication; dashboard; upload; project system; image preview; AI enhancement; upscaling; background removal; basic vector conversion; halftone generator; embroidery visual effect; image resizing; DPI checker; transparency checker; PNG/SVG/PDF export; responsive UI.

SHOULD HAVE

AI Artwork Inspector; before/after comparison; processing history; basic DTF preview.

NOT IN MVP

Marketplace; complex subscriptions/billing; direct printer control; RIP integration; full embroidery digitization; native mobile app; AI image generation; team collaboration.

30. Success Metrics

Primary

A real user should be able to upload a problematic image, identify issues, improve it, prepare it for DTF, and export it without requiring Photoshop or Illustrator for the basic workflow.

Processing target

Standard artwork should generally move from upload to export within approximately five minutes, subject to file size and AI workload.

Quality

The system should identify a high proportion of common DTF artwork problems in a controlled test dataset.

31. Product Roadmap

Phase 1 — Foundation

Architecture, authentication, dashboard, project system, upload, editor, storage and API.

Phase 2 — AI Engine

Enhancement, upscaling, background removal and AI Inspector.

Phase 3 — Creative Engine

Vector, halftone, embroidery and effects.

Phase 4 — DTF Engine

DTF inspection, underbase preview, print dimensions and gang sheets.

Phase 5 — Commercial

Subscriptions, licensing, usage limits, billing and analytics.

Phase 6 — Pulse Ecosystem

Printer profiles, RIP integrations, training, cloud projects, customer portal and marketplace.

32. Definition of Done — MVP

A complete MVP must allow a user to:

Create an account → create a project → upload poor-quality artwork → analyze it → improve it → remove its background → apply an effect → check DTF suitability → export → reopen the project later.

33. Commercial Product Guardrails

No unrealistic guarantee

The product should be positioned as AI-powered artwork preparation and quality assistance—not a guarantee that every design will print perfectly.

Reason

Final print results also depend on printer hardware, RIP software, inks, film, powder, heat-press settings, garment material, and operator settings.

Brand position

Pulse Artwork Studio AI should strengthen Pulse Machinery's broader customer ecosystem: hardware, software, training, support, and future cloud services.

Approval & Next Step

This PRD establishes the product boundary and requirements for Pulse Artwork Studio AI Version 1.0. The next project document should translate these requirements into an implementation blueprint.

Next document: Pulse Artwork Studio AI — System Architecture & Technical Design Specification v1.0

That document will define the repository structure, Flutter architecture, backend services, database schema, API endpoints, authentication flow, processing-job architecture, object storage model, AI service boundaries, security model, deployment strategy, development environment, and testing strategy.